



SETUP GUIDE

Minikube + kubectl Local Kubernetes Environment

Tools: Minikube (local K8s cluster) + kubectl (K8s CLI)

Prerequisite: Docker Desktop running (installed in Week 4)

SECTION 1: WSL (Windows Subsystem for Linux) — RECOMMENDED FOR WINDOWS

PREREQUISITES

- Windows 10/11 with WSL2 enabled (from Week 4)
- Docker Desktop running and integrated with WSL2
- At least 4 GB free RAM (Minikube default allocation: 2 GB)
- At least 20 GB free disk space

STEP 1: Install kubectl

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s  
https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"
```

```
sudo install -o root -g root -m 0755 kubectl /usr/local/bin/kubectl
```

```
rm kubectl
```

Verify:

```
kubectl version --client
```

Expected output: Client Version: v1.XX.X (version number)

STEP 2: Install Minikube

```
curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
```

```
sudo install minikube-linux-amd64 /usr/local/bin/minikube
```

```
rm minikube-linux-amd64
```

Verify:

```
minikube version
```

Expected output: minikube version: v1.XX.X



STEP 3: Start Minikube (uses Docker driver — recommended)

```
minikube start --driver=docker
```

First run: downloads Kubernetes components (2–5 minutes)

Subsequent runs: 30–60 seconds

Expected output:

- ✓ minikube v1.XX.X
- ✓ Using the docker driver
- ✓ Starting control plane node
- ✓ kubectl is now configured to use "minikube" cluster

STEP 4: Verify Cluster

```
kubectl get nodes
```

Expected:

NAME	STATUS	ROLES	AGE	VERSION
minikube	Ready	control-plane	1m	v1.XX.X

```
kubectl get pods -A
```

Expected: several pods in kube-system namespace, all Running or Completed

STEP 5: Quick Deployment Test

```
kubectl create deployment hello-k8s --image=nginx:latest
```

```
kubectl get pods # Wait for STATUS: Running
```

```
kubectl expose deployment hello-k8s --type=NodePort --port=80
```

```
minikube service hello-k8s --url
```

Open the URL in browser → should see nginx welcome page.

Clean up:

```
kubectl delete deployment hello-k8s
```

```
kubectl delete service hello-k8s
```

SECTION 2: macOS

PREREQUISITES

- macOS 12+ (Monterey or later recommended)
- Docker Desktop running



- Homebrew installed
- At least 4 GB free RAM
- At least 20 GB free disk space

STEP 1: Install kubectl

brew install kubectl

Verify:

kubectl version --client

STEP 2: Install Minikube

brew install minikube

Verify:

minikube version

STEP 3: Start Minikube

minikube start --driver=docker

(Same as WSL — uses Docker Desktop as the driver)

STEP 4: Verify Cluster

kubectl get nodes # Should show minikube node, STATUS: Ready

kubectl get pods -A # System pods running

STEP 5: Quick Deployment Test

kubectl create deployment hello-k8s --image=nginx:latest

kubectl get pods

kubectl expose deployment hello-k8s --type=NodePort --port=80

minikube service hello-k8s --url

Open URL in browser

Clean up:

kubectl delete deployment hello-k8s

kubectl delete service hello-k8s



SECTION 3: LINUX (Ubuntu/Debian)

PREREQUISITES

- Ubuntu 20.04+ or Debian 11+
- Docker Engine installed and running (from Week 4)
- Current user in docker group (sudo usermod -aG docker \$USER)
- At least 4 GB free RAM
- At least 20 GB free disk space

STEP 1: Install kubectl

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s
```

```
https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"
```

```
sudo install -o root -g root -m 0755 kubectl /usr/local/bin/kubectl
```

```
rm kubectl
```

Verify:

```
kubectl version --client
```

STEP 2: Install Minikube

```
curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
```

```
sudo install minikube-linux-amd64 /usr/local/bin/minikube
```

```
rm minikube-linux-amd64
```

Verify:

```
minikube version
```

STEP 3: Start Minikube

```
minikube start --driver=docker
```

Note: Do NOT run minikube as root. Run as regular user with Docker access.

STEP 4: Verify Cluster

```
kubectl get nodes
```

```
kubectl get pods -A
```

STEP 5: Quick Deployment Test

```
kubectl create deployment hello-k8s --image=nginx:latest
```

```
kubectl get pods
```



```
kubectl expose deployment hello-k8s --type=NodePort --port=80
```

```
minikube service hello-k8s --url
```

Clean up:

```
kubectl delete deployment hello-k8s
```

```
kubectl delete service hello-k8s
```

SECTION 4: USEFUL MINIKUBE COMMANDS

DAILY WORKFLOW

minikube start	Start cluster (or resume stopped cluster)
minikube status	Check if cluster is running
minikube stop	Stop cluster (preserves state and deployments)
minikube delete	Delete cluster completely (fresh start)

RESOURCE MANAGEMENT

minikube start --cpus=4 --memory=4096	Start with more resources
minikube dashboard	Open K8s web dashboard in browser
minikube addons list	Show available addons
minikube addons enable metrics-server	Enable metrics (for kubectl top)

KUBECTL ESSENTIALS

kubectl get pods	List pods in default namespace
kubectl get pods -A	List ALL pods (all namespaces)
kubectl get services	List services
kubectl get deployments	List deployments
kubectl describe pod <name>	Detailed pod info + events
kubectl logs <pod-name>	View pod logs
kubectl top pods	Resource usage (needs metrics-server)



`kubectl delete pod <name>` Delete a pod (ReplicaSet recreates it)

SECTION 5: TROUBLESHOOTING

PROBLEM: "minikube start" fails with Docker driver error

CAUSE: Docker is not running or not accessible

FIX:

- Ensure Docker Desktop is running (WSL/macOS)
- Ensure Docker Engine is running: `sudo systemctl start docker` (Linux)
- Ensure user is in docker group: `groups | grep docker`
- Restart Docker if needed

PROBLEM: `kubectl` commands return "connection refused"

CAUSE: Minikube cluster is not running

FIX:

`minikube status` # Check status

`minikube start` # Start if stopped

PROBLEM: Pod stuck in "Pending" state

CAUSE: Insufficient resources on node

FIX:

`kubectl describe pod <name>` # Check Events section for reason

`minikube stop`

`minikube start --cpus=4 --memory=4096` # Allocate more resources

PROBLEM: Pod stuck in "ImagePullBackOff"

CAUSE: Cannot pull container image (typo or network issue)

FIX:

`kubectl describe pod <name>` # Check image name in Events

Verify image name and tag are correct

Check internet connectivity

PROBLEM: "minikube service" returns no URL (WSL specific)



CAUSE: WSL networking isolation

FIX:

```
minikube service hello-k8s --url
```

If URL not accessible from Windows browser, use:

```
kubectrl port-forward service/hello-k8s 8080:80
```

Then open <http://localhost:8080> in Windows browser

PROBLEM: Minikube using too much disk space

FIX:

```
minikube delete          # Remove cluster
```

```
docker system prune -a    # Clean Docker cache
```

```
minikube start            # Fresh start
```

PROBLEM: kubectrl version mismatch warning

CAUSE: kubectrl version differs from cluster version by >1 minor

FIX: Update kubectrl to latest:

WSL/Linux:

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s  
https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectrl"
```

```
sudo install -o root -g root -m 0755 kubectrl /usr/local/bin/kubectrl
```

macOS:

```
brew upgrade kubectrl
```

SECTION 6: VERIFICATION CHECKLIST

Run these commands and confirm expected outputs:

```
[ ] kubectrl version --client
```

→ Shows client version (no errors)

```
[ ] minikube version
```



→ Shows minikube version

[] minikube start --driver=docker

→ Cluster starts successfully

→ "kubectl is now configured to use minikube cluster"

[] kubectl get nodes

→ 1 node, STATUS: Ready

[] kubectl get pods -A

→ System pods in kube-system namespace, all Running/Completed

[] kubectl create deployment test --image=nginx

→ deployment.apps/test created

[] kubectl get pods (wait 30 sec)

→ test-XXXXXX pod, STATUS: Running

[] kubectl expose deployment test --type=NodePort --port=80

→ service/test exposed

[] minikube service test --url (or kubectl port-forward)

→ URL accessible, nginx page loads

[] kubectl scale deployment test --replicas=3

→ 3 pods running (kubectl get pods shows 3)

[] kubectl delete deployment test && kubectl delete service test

→ Resources cleaned up

ALL CHECKS PASSED = Environment ready for Week 5 exercises